4

Context

Introduction to the idea that math is all relative to context as things behave differently in different contexts. A little more formality.

Mathematics is often thought of as being rigid and fixed, having "absolute truth", clear right and wrong, and unbreakable laws. In this chapter we are going to examine a sense in which this is an unnecessarily and unrealistically narrow view of mathematics. Like many stereotypes it has a kernel of truth to it, but that small truth has been blown out of proportion.

We are going to look at how mathematical objects behave very differently in different contexts; thus they have no fixed characteristics, just different characteristics in different contexts. Thus truth is not absolute but is contextual, and so we should always be clear about the context we're considering. This idea is central to category theory.

Pedantically one might declare that the "truth-in-context" is then absolute, but I think this amounts to saying that truth is *relative* to context. Your preferred wording is a matter of choice, but I have made my choice because I think it is important to focus our attention on the context in which we are working, and not to regard anything as fixed.

Moreover, I don't think "truth-in-context is absolute" is what anyone typically means by the "rigidity" of mathematics; usually they're not thinking about different things being true in different contexts. For example, people who aren't mathematicians often declare that one plus one just *is* two. But we are going to see that even this basic "truth" is relative to context.

There *is* a certain rigidity to mathematics, but there's also a crucial fluidity, and the two work together. Fluidity comes from the vast universe of mathematical worlds that we can move between. Rigidity or constraint comes from a decision to move between worlds only in a way that makes sense, like pouring water into a glass without spilling it, or going to the moon without destroying your space ship. This is a different sort of constraint from the kind where you don't go anywhere at all and, moreover, you assume there's nowhere to go.

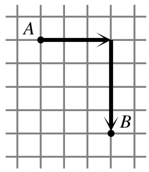
The popular concept of math is that it is a dead tree, completely fixed and dry. I think a more accurate view is that it's a living tree, whose base is fixed but it can still sway and its branches and roots still grow. An impatient explorer might say "Trees are boring. They don't move." And yes, perhaps compared with lions and elephants they don't move. But if you take time to stare at them long enough they might become fascinating.

4.1 Distance

We're going to look at a world in which some familiar things behave differently from usual. Although actually it's a very common world in "real life", it just behaves differently from some common mathematical worlds that might be regarded as "fixed", which is not really *its* fault.

We're going to think about being in a city with streets laid out on a grid — so probably not a European city. Of course, even American cities aren't on a *perfect* grid — there are usually some diagonal streets somewhere, but we're going to imagine a perfectly regular grid with evenly spaced parallel roads.

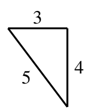
Now, when traveling from A to B we can't take the diagonal because there isn't a street there. Instead, the distance we'd actually have to go along the streets will be something like in this diagram. We have to go 3 blocks east and 4 blocks south making a total of 7 blocks.



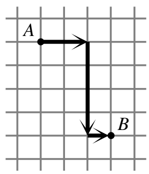
Calculating the distance "as the crow flies", that is, the direct distance through the air in a straight line regardless of obstacles, is rather academic as we can't usually make use of that route (unless we're pinpointing a location by sound or something, as in the film "Taken 2").

Things To Think About

T 4.1 We can use Pythagoras to calculate the distance as the crow flies which in this case will be 5 blocks. Is the road distance always more than the crow distance?

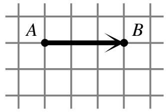


Is the road distance between two places always the same even if we turn at different points, for example as in this diagram?

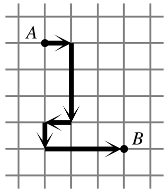


This type of distance is sometimes called "taxi-cab" distance (as if we all travel around in taxis all the time).

The taxi distance can be the same as the crow distance if A and B are on the same street, but the taxi distance can never be *smaller* than the crow distance.



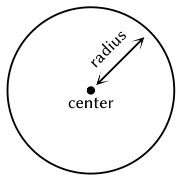
Also it doesn't matter where you turn as long as you don't go back on yourself, because wherever you turn you still have to cover the same number of blocks east and the same number of blocks south, making the same total. By contrast the path shown on the right covers more blocks because it does go back on itself.



It might seem like I'm trying to say something complicated, but it really is just like counting the blocks you actually walk when you walk around a city laid out on a grid. If your brain is flashing "Math class!" warning lights then you might be trying to read too much into this. No hypotenuse is involved.

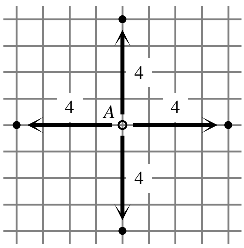
We can now think about what a circle looks like in this world. You might think a circle is just a familiar shape, but there is a *reason* that shape looks like that, and in math we are interested in reasons, or ways we can characterize things precisely.

How could you describe a circle to someone down the phone? One way is to say that you pick a center and a distance and draw all the points that are this distance away from that center. That chosen distance is called the *radius*.

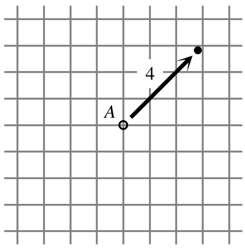


But in the taxi world we can't do distance along diagonals, so what will "all the points the same distance from a chosen center" look like? Let's try a circle of radius 4 blocks.

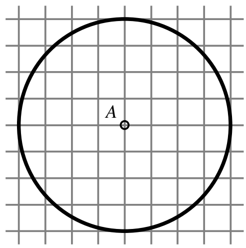
Here are the most obvious points that are a distance of 4 blocks from A.



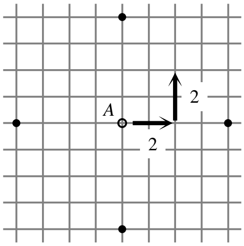
Here is something we can't do.



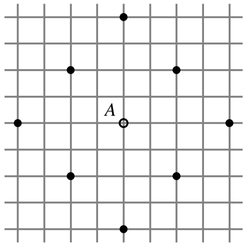
So we definitely can't get all these points — most aren't on the grid at all.



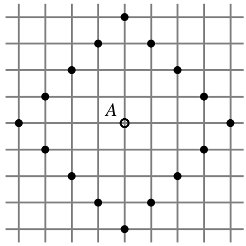
However we can go 4 blocks with a turn in the middle.



If we do this in all possible directions we get these points.

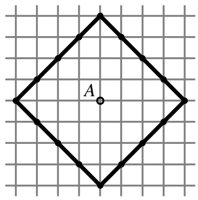


If we also do 3 blocks and 1 block to make 4 we get all these points.



The last picture is what a circle looks like in this taxi world. After filling in the points for going “2 blocks and 2 blocks” you might have seen the pattern to help you realize you could also do 3 and 1, which is good.

However, you might also have been tempted to join the dots like this picture. You are welcome to do so on paper but it won’t mean anything in the taxi world because those lines are not lines we can travel on. They include points we can’t get to in the taxi world — the taxi world circle really doesn’t include those lines.



So a circle in this taxi world is just a collection of disconnected dots in a diamond shape. How does that make you feel? Do you feel uncomfortable, as if this somehow violates natural geometry? Or do you feel tickled that a circle can look so funny? Both reactions are valid. The main thing is to appreciate that even some of the most basic things we think we know are only true in a particular context, and things can look very different in another context. It is important to be

a) clear what context we’re considering at any given moment (which we usually aren’t in basic math lessons), and
b) open to shifting context and finding different things that can happen.

Technicalities

What we have done here is find a different scheme for measuring distance between points in space. In fact there are many different possible ways of measuring distance and these are called *metrics*. Not every scheme for measuring distance will be a reasonable one, and in order to study this and any abstract concept rigorously we decide on criteria for what should count as reasonable. A metric space is then a set of points endowed with a metric. The idea of a metric space is to focus our attention on not just the points we're thinking about, but the *type of distance* we're thinking about.

The usual way of measuring distance "as the crow flies" is called the Euclidean metric. The taxi-cab way really is called the taxi-cab metric or the rectilinear metric. More formally it is called the L_1 metric and is the first in an infinite series of L_n metrics. The next one, L_2, is in fact the Euclidean metric.

4.2 Worlds of numbers

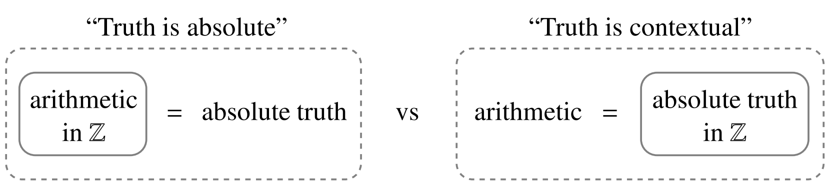
Many people say to me "Well, one plus one just *does* equal two." I reply, "In some worlds it's zero."

It's true that $1 + 1 = 2$ in ordinary numbers. But that's because it's how "ordinary numbers" are defined. Most people who think $1 + 1$ just *does* equal 2 are not considering that this is only true in some contexts and not others, as they're so used to one particular context. This is a bit like people who've never visited another country, and don't realize that people drive on the other side of the road in other places. Some people who haven't traveled don't understand that some ways of doing things are highly cultural and possibly arbitrary. For example:

- "it's math not maths" (not in the UK);
- "steering wheels are on the right of a car" (in the UK).

Some people can't imagine not having a car and others can't imagine having one.

We have seen that distance is contextual and thus "circles" are also dependent on context. We will now see a way that the behavior of numbers is also dependent on context. So all the arithmetic we are forced to learn in school is contextual, not fixed; it's not an absolute truth of the universe, unless we take its context as part of that truth. The context is the integers, that is, all the whole numbers: positive, negative and zero. The set of integers is often written as $\mathbb{Z}$. Here is a diagram showing those different points of view.



Arithmetic might seem like absolute truth if you think the integers are the only possible context. But in fact most people *do* know other contexts, they just don't come to mind when thinking about arithmetic. For example if you dump a pile of sand onto a pile of sand you still just get one pile of sand; it's just bigger.

Things To Think About

T 4.2 What contexts can you think of in which $1 + 1$ is something other than 2? Can you think of other contexts in which it's 1? What about 0, or 3 or more? What about other ways in which arithmetic sometimes works differently?

Here are some places where arithmetic works differently.

Telling the time. 2 hours later than 11 o'clock isn't 13 o'clock unless you're using a 24-hour clock. On a 12-hour clock it's 1 o'clock, that is $11 + 2 = 1$. On a 24-hour clock 2 hours later than 23 o'clock is not 25 o'clock, it's 1 o'clock, that is $23+2 = 1$. (While we don't usually say "23 o'clock" out loud in English, it does happen in French.)

I'm not not hungry. Particular kinds of children find it amusing to say things like "I'm not not hungry" to mean "I am hungry". If we count the instances of "not" we get $1 + 1 = 0$.

Rotations. If you rotate on the spot by one quarter-turn four times in the same direction you get back to where you started, as if you had done zero quarter-turns. So if we count the quarter-turns, $1 + 1 + 1 + 1 = 0$. We could generalize this to any n by rotating n times by $\frac{1}{n}$ of a turn each time.

Mixing paint. If you add one color paint to another you do not get two colors, you get one color.[†] Likewise a pile of sand or drop of water. So $1 + 1 = 1$.

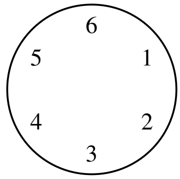
Pairs. If one pair of tennis players meets up with another pair for an afternoon of tennis, there are 6 potential pairs of tennis players among them, if everyone is happy to partner with any of the others.[‡]

The first three of these situations all have something in common and we are now going to express exactly what that analogy might be.

[†] This example was brought to my attention by my art students at SAIC.

[‡] This example was also brought to my attention by my art students at SAIC, though in a less child-friendly formulation.

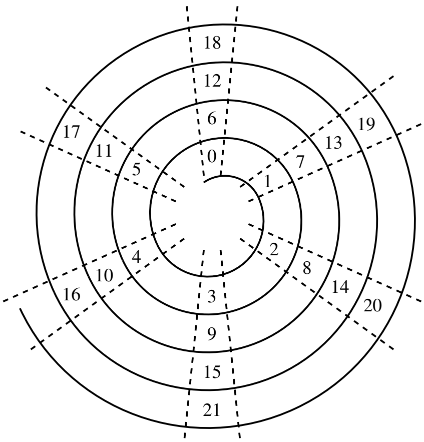
In each of those first three cases there is one fixed number n that is like 0, in the sense that once you get to n you start counting again from 0 like going round a clock. In the first example n is 12 (or 24), in the next one it is 2, and in the third it is 4 and then n. We could depict this as a generalized clock with n hours. Here is a 6-hour clock, which would be relevant if, say, you had to take a medicine every 6 hours.



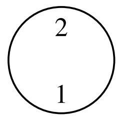
On this clock we have various relationships as in the table below, but we could also draw them on a spiral as if we had taken an ordinary number line and wrapped it around itself:

7	is the same as	1
8	is the same as	2
9	is the same as	3
	⋮	

Each of the six positions on the clock has a whole tower of numbers that "live" there.



We could make an analogous clock for any fixed number n. Here is a 2-hour clock:



These pictures are vivid but not very practical for working anything out. Imagine trying to work out where 100 sits on the 6-hour clock. You could count all the way round the clock to 100 but it would be slow, tedious and boring.

Things To Think About

T 4.3 Can you see a relationship between all the numbers living in the same place? This would give us a *relative* characterization of all the numbers that count as the same. Can you also think of a *direct* characterization?

This is just a glimpse of the world of n-hour clocks, which are technically called the "integers modulo n". In Chapter 6 will discuss why we might want to represent this world more formally, and how we do it. The idea is that it makes it easier to work out exactly what is going on and be sure we are right.

Abstract math looks at more and more abstract accounts of these worlds that make connections with more, apparently unrelated, structures.†

For now, however, we just need to appreciate that this is a different context in which numbers interact and behave differently from usual.

4.3 The zero world

When I was little, I was allergic to artificial food color. In those days all candy had it, which meant that no matter how many sweets I was given, I effectively had 0. I lived in the zero world of candy, in which everything equals 0. This is the world you end up in if you decide to try and declare $1+1 = 1$, but still want some other usual rules of arithmetic to hold: in that case we could subtract 1 from both sides and get $1 = 0$. If that's true then *everything* will be 0. (You might like to try proving that.)

By contrast, in the world of paint we have $1 + 1 = 1$ without landing in the zero world — if we add 1 color to 1 color we get 1 color, and it is not the same as having 0 colors. The difference here is that colors do not obey the other usual rules of arithmetic. In particular you can't subtract colors and so you can't "subtract 1 color from both sides" as we did in the argument involving numbers. This is how that situation avoids being in the zero world.

The zero world might not seem very interesting — you can't really do anything in it. Or rather, you can do *anything* in it and it's all equally valid. It turns out that a world in which everything is equally valid is not very interesting, and is more or less the same as a world in which nothing is possible.

In fact, while this world is not very interesting, it is very important, like some people who are important but not very interesting. We will later see that this world is a useful one to have in the universe. So inside the context of the zero world itself it's not that interesting, but if we zoom out to the context of the universe, the zero world is important and useful. It's another case where context makes a difference.

The zero world is important in the universe because of its extreme relationships with everything else in the universe. In the next chapter we're going to see how different contexts can arise from considering different types of relationship.

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† They are examples of quotient groups in group theory, which are examples of coequalizers in category theory. I say this just in case you feel like looking them up.